

Kernel experiments for lemma validation on S^1

Shreyas Kalvankar

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1 Finite-sample lemma validation for the analytic NTK on S^1

1.1 Hypotheses and objects

We work on the circle S^1 , parameterized by $\gamma \in [0, 2\pi)$, with uniform measure

$$\mu(d\gamma) = \frac{d\gamma}{2\pi}.$$

We consider the analytic circle NTK $\Theta(\gamma, \psi) = \Theta(\gamma - \psi)$, so the associated continuum operator is

$$(Tf)(\gamma) = \int_0^{2\pi} \Theta(\gamma, \psi) f(\psi) \mu(d\psi).$$

Because Θ is translation-invariant on the circle, the real Fourier modes diagonalize T . We use

$$\varphi_0(\gamma) = 1, \quad \varphi_{k,c}(\gamma) = \sqrt{2} \cos(k\gamma), \quad \varphi_{k,s}(\gamma) = \sqrt{2} \sin(k\gamma), \quad k \geq 1.$$

For each frequency k , the corresponding continuum eigenvalue is denoted by λ_k , so

$$T\varphi_0 = \lambda_0\varphi_0, \quad T\varphi_{k,c} = \lambda_k\varphi_{k,c}, \quad T\varphi_{k,s} = \lambda_k\varphi_{k,s}.$$

We then sample

$$\phi_1, \dots, \phi_n \stackrel{\text{iid}}{\sim} \text{Unif}[0, 2\pi),$$

and form the empirical operator

$$(A_nv)_i = \frac{1}{n} \sum_{j=1}^n \Theta(\phi_i, \phi_j) v_j.$$

Equivalently, if $K_{ij} = \Theta(\phi_i, \phi_j)$, then

$$A_n = \frac{1}{n} K.$$

We choose a fixed low-frequency cutoff $\mathbf{K}_{\max}$, and evaluate the Fourier basis on the sampled points. This gives the matrix

$$\Phi \in \mathbb{R}^{n \times d}, \quad d = 2\mathbf{K}_{\max} + 1.$$

where Φ is the matrix $\left(\varphi_0(\gamma_i) \quad \varphi_{1,c}(\gamma_i) \quad \varphi_{1,s}(\gamma_i) \quad \dots \quad \varphi_{k,c}(\gamma_i) \quad \varphi_{k,s}(\gamma_i) \right)_{i=1}^n$

1.2 Finite-sample objects and hypotheses

The key finite-sample objects are the following.

1. Sampled Fourier geometry. We define

$$G = \frac{1}{n} \Phi^\top \Phi.$$

This is the sampled Gram matrix of the low-frequency Fourier basis. It measures whether the sampled Fourier vectors remain approximately orthonormal. Our first hypothesis is that

$$G \approx I.$$

Equivalently, the sampled Fourier geometry should concentrate near the continuum orthonormal geometry.

2. Compressed empirical operator. We define

$$H = \frac{1}{n} \Phi^\top A_n \Phi.$$

This is the empirical operator compressed onto the sampled low-frequency Fourier block. Since the continuum operator is diagonal in the Fourier basis, the natural hypothesis is that H should be approximately diagonal as well.

3. Approximate eigenvalues. For each mode of frequency k , define

$$\lambda_k^{(n)} = \left(1 - \frac{1}{n}\right) \lambda_k + \frac{1}{n} g(0), \quad g(0) = \Theta(x, x) = \Theta(0).$$

These are the finite-sample approximate eigenvalues.

The motivation for this definition is that for a sampled Fourier mode φ_k ,

$$\mathbb{E} \left[\frac{1}{n} \varphi_k^\top A_n \varphi_k \right] = \left(1 - \frac{1}{n}\right) \lambda_k + \frac{1}{n} g(0).$$

Thus $\lambda_k^{(n)}$ is the expected diagonal action of the empirical operator on that sampled mode.

4. Approximate eigen-action error. We define

$$E = A_n \Phi - \Phi \Lambda^{(n)},$$

where $\Lambda^{(n)}$ is the diagonal matrix whose entries are the $\lambda_k^{(n)}$, repeated for cosine and sine at the same frequency k .

This measures how close the sampled Fourier basis vectors are to actual empirical eigenvectors.

With these objects in place, the three finite-sample hypotheses can be stated as follows.

Hypothesis A: concentration of the sampled Fourier geometry.

$$G = \frac{1}{n} \Phi^\top \Phi \approx I.$$

Interpretation: the sampled Fourier vectors remain almost orthonormal.

Hypothesis B: concentration of the compressed empirical operator.

$$H = \frac{1}{n} \Phi^\top A_n \Phi \approx \Lambda^{(n)}.$$

Interpretation: inside the low-frequency Fourier block, the empirical operator is approximately diagonal, with diagonal entries given by the approximate eigenvalues.

Hypothesis C: approximate eigen-action of the sampled Fourier modes.

$$A_n \Phi \approx \Phi \Lambda^{(n)}.$$

Interpretation: each sampled Fourier mode is acted on approximately by scalar multiplication with $\lambda_k^{(n)}$.

These are the three finite-sample lemmas in operational form.

1.3 What is expected to concentrate

More explicitly, as n grows, we expect:

$$\|G - I\| \rightarrow 0,$$

$$\|H - \Lambda^{(n)}\| \rightarrow 0,$$

and

$$\|A_n \Phi - \Phi \Lambda^{(n)}\| \rightarrow 0.$$

Mode-by-mode, this means

$$\frac{\|A_n \varphi_k - \lambda_k^{(n)} \varphi_k\|_2}{\|\varphi_k\|_2} \rightarrow 0$$

for low frequencies k , with cosine and sine treated separately.

1.4 What this gets us dynamically

Now consider kernel gradient descent with zero initialization:

$$y_{t+1} = y_t - \eta A_n r_t, \quad r_t = y_t - y^*.$$

Equivalently,

$$r_{t+1} = (I - \eta A_n) r_t.$$

If the sampled Fourier modes were exact eigenvectors of A_n , then the residual coefficient along a mode of frequency k would evolve as

$$\alpha_k(t+1) = (1 - \eta \lambda_k^{(n)}) \alpha_k(t),$$

hence

$$\alpha_k(t) = (1 - \eta \lambda_k^{(n)})^t \alpha_k(0).$$

So the concentration of G , H , and E gives the following consequence: the residual dynamics should approximately decouple along low Fourier modes, and each low mode should decay at a rate determined by its approximate eigenvalue $\lambda_k^{(n)}$.

That is the central reason we care about the three static operator lemmas.

1.5 Target

The target

$$f^*(\gamma) = 1 + 0.9 \cos(\gamma) + 0.5 \cos(2\gamma) + 0.35 \cos(3\gamma)$$

lives only in the first few Fourier modes.

This is useful because:

- it makes the mode-wise decay easy to interpret,
- it isolates the low-frequency part where the concentration is expected to be best,
- and any nonzero coefficients outside the target frequencies can be interpreted as finite-sample leakage.

1.6 Operator concentration under uniform sampling

We first examine the three static finite-sample concentration statements:

$$G = \frac{1}{n} \Phi^\top \Phi \approx I, \quad H = \frac{1}{n} \Phi^\top A_n \Phi \approx \Lambda^{(n)}, \quad A_n \Phi \approx \Phi \Lambda^{(n)}.$$

For each quantity, we plot the maximum-entry error against the sample size n , aggregating over multiple random seeds. The solid blue line shows the mean over seeds, the shaded band shows one standard deviation, and the individual points show the per-seed values. In addition, the dashed orange curve shows the theoretical concentration scale predicted by the finite-sample lemmas.

Why these are the correct theoretical rates. The lemmas give exponential tail bounds in the deviation parameter ε at fixed sample size n , rather than exponential decay of the plotted error as a function of n . More precisely, the note gives bounds of the form

$$\mathbb{P}(\|G - I\|_{\max} \geq \varepsilon) \leq 2d^2 e^{-n\varepsilon^2/8},$$

$$\mathbb{P}(\|H - \mathbb{E}H\|_{\max} \geq \varepsilon) \leq 2d^2 e^{-n\varepsilon^2/(32\kappa^2)},$$

and

$$\mathbb{P}(\|A_n \Phi - \Phi \Lambda^{(n)}\|_{\max} \geq \varepsilon) \leq 2nd e^{-n\varepsilon^2/(4\kappa^2)}.$$

Solving these bounds for the deviation size ε at confidence level $1 - \delta$ yields the concentration scales

$$\varepsilon_G(n, \delta) \asymp \sqrt{\frac{\log(2d^2/\delta)}{n}}, \quad \varepsilon_H(n, \delta) \asymp \kappa \sqrt{\frac{\log(2d^2/\delta)}{n}},$$

and

$$\varepsilon_E(n, \delta) \asymp \kappa \sqrt{\frac{\log(2nd/\delta)}{n}}.$$

Thus the expected rates are:

$$\|G - I\|_{\max} \sim n^{-1/2}, \quad \|H - \Lambda^{(n)}\|_{\max} \sim n^{-1/2}, \quad \|A_n \Phi - \Phi \Lambda^{(n)}\|_{\max} \sim \sqrt{\frac{\log n}{n}},$$

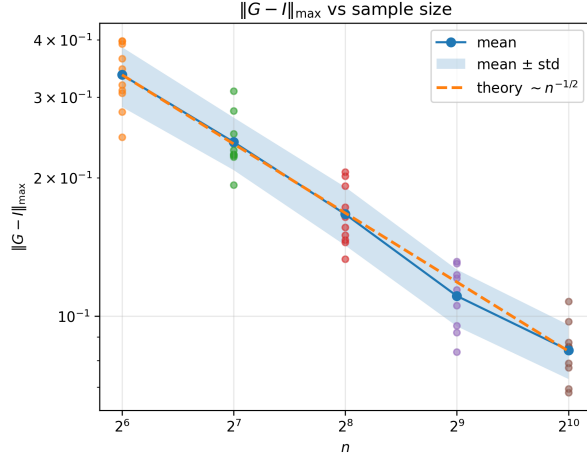
up to constants and lower-order logarithmic factors. For the bias kernel, a natural constant choice is

$$\kappa = \sup_{\theta} |\Theta(\theta)| = \Theta(0) = \frac{3}{2},$$

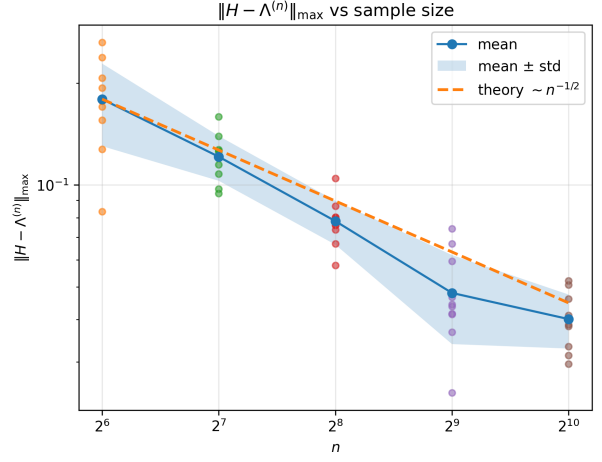
since the kernel is nonnegative and is maximized on the diagonal.

Observations. Figure 1 shows a clear and consistent concentration trend across all three operator-level diagnostics.

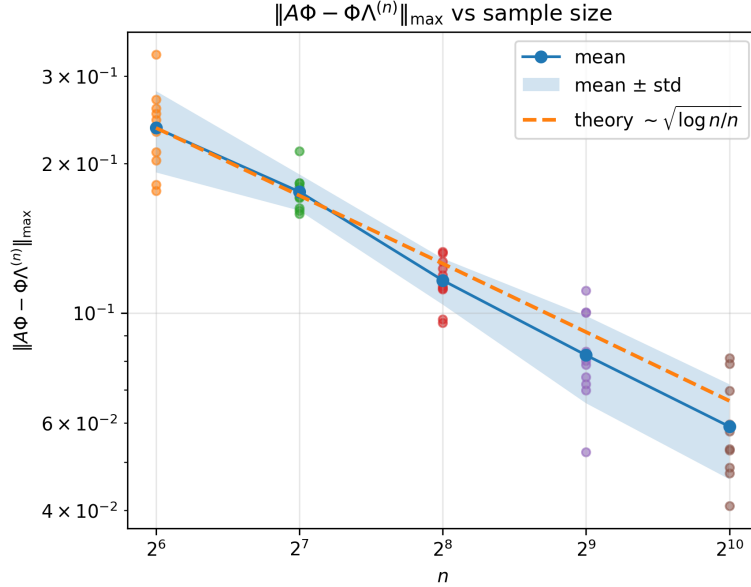
- The sampled Gram matrix error $\|G - I\|_{\max}$ decreases steadily with n and closely follows the predicted $n^{-1/2}$ behavior. This indicates that the sampled low-frequency Fourier vectors become increasingly close to an orthonormal system as the number of sampled points grows.
- The compressed operator error $\|H - \Lambda^{(n)}\|_{\max}$ also decreases with n , again in a way consistent with the predicted $n^{-1/2}$ scaling. This supports the claim that, inside the sampled low-frequency Fourier block, the empirical operator becomes increasingly close to a diagonal operator with entries given by the approximate eigenvalues.
- The action error $\|A_n \Phi - \Phi \Lambda^{(n)}\|_{\max}$ decreases more slowly, consistent with the extra logarithmic factor coming from the stronger union bound in the corresponding lemma. Empirically, the plotted mean tracks the predicted $\sqrt{\log n / n}$ scale well.



(a) Gram concentration: $\|G - I\|_{\max}$ versus n .



(b) Compressed operator concentration: $\|H - \Lambda^{(n)}\|_{\max}$ versus n .



(c) Approximate eigen-action concentration: $\|A_n \Phi - \Phi \Lambda^{(n)}\|_{\max}$ versus n .

Figure 1: Finite-sample operator concentration for the sampled Fourier geometry, compressed operator, and approximate eigen-action. The dashed orange curves indicate the theoretical concentration scales predicted by the lemmas: $n^{-1/2}$ for the Gram and compressed operator errors, and $\sqrt{\log n/n}$ for the action error. In all three cases, the empirical mean decreases with n and tracks the expected scaling well.